Tidal potential measured with distant optical clocks

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The Moon and the Sun change Earth's gravitational potential directly and through solid-Earth deformation and ocean motion,[1,2] altering the relative rates of distant clocks.[3] Fibre-linked optical clocks can track these changes,[4,5] but measuring tidal signals and quantitatively separating their overlapping components remain challenging. Here we detect tidal variations in the potential difference between a strontium clock[6] in coastal Shanghai and an ytterbium clock[7] in inland Wuhan, 692 km apart. Across 545.9 hours of observations spanning 353 days, the response to the combined solid-Earth and ocean-loading prediction is 0.96 ± 0.11. The lunar semidiurnal group has a response of 1.00 ± 0.13. Spectral peaks near the M_2 and S_2 frequencies occur at 22.361 and 23.148 μHz, with equivalent-height amplitudes of 4.87 and 2.87 cm, respectively. Each contribution remains identifiable after controlling for the other. We develop a signal-to-noise model that combines signal strength, clock noise and actual sampling, accounts for the lunar precision, and helps explain the weaker constraints on other components. These results help characterize the tidal background in optical-clock networks, with potential applications to the redefinition of the second[8] and the unification of height systems at the centimetre level.[9]

Introduction

Optical clocks connected by frequency-transfer links measure gravitational potential differences through the gravitational redshift. In the weak-field limit of general relativity, a potential difference ΔU produces a fractional frequency shift $\Delta\nu/\nu \simeq \Delta U/c^2$, where c is the speed of light.[3] Evaluated fractional systematic uncertainties now approach 10^{-19}.[10] Near Earth's surface, fractional shifts of 10^{-18}, 10^{-19} and 10^{-20} correspond to equivalent heights of approximately 1 cm, 1 mm and 0.1 mm.[5,11] The redshift has been resolved within a millimetre-sized atomic sample,[11] while fibre networks and free-space frequency transfer extend comparisons to distant sites.[4,12–14] Proposed quantum clock networks[15] have also reached an elementary demonstration with entangled clocks.[16]

Clock comparisons complement levelling and satellite gravity observations.[17,18] Earth's gravitational potential also changes with volcanic activity,[19] tsunamis[20] and water storage.[21] These contributions overlap in a clock record, so detecting a weak signal requires identifying the stronger variations and knowing how accurately each can be removed. Tides provide a predictable signal with which to develop this capability. Lunar and solar contributions must be distinguished from each other and from laboratory variations that follow the solar day, while observational gaps spread spectral power into neighbouring frequencies. Subtracting the tidal background therefore requires separating the tidal components and their sampling sidelobes and quantifying the precision of each component.

Here we report measurements of tidal changes in the potential difference between a strontium clock in coastal Shanghai[6] and an ytterbium clock in inland Wuhan,[7] 692 km apart and connected by a 1,350-km fibre link.[22] Over 545.9 hours of observations spanning 353 days, we measure the combined solid-Earth and ocean-loading tide and separately detect the principal lunar semidiurnal contribution (M_2) and the contribution at the principal solar semidiurnal frequency (S_2). The two contributions drift in relative phase and produce neighbouring spectral peaks, and each remains identifiable after the other is removed. We develop a signal-to-noise model that combines signal strength, noise and the actual sampling, and show that it accounts for most of the differences in precision between components.

Results

A coastal–inland clock baseline

As shown in Fig. 1a, the Shanghai Sr clock and the Wuhan Yb clock lie on a 692-km coastal–inland baseline and are connected through Hefei by a 1,350-km phase-stabilized fibre link.[22] The comparison instability is approximately $5.4 \times 10^{-16}/\sqrt{\tau/\mathrm{s}}$ at averaging time τ for the spliced summer record (ref. 23) and is dominated by the clocks. Changes in the frequency ratio follow the changing potential difference, and a fitted offset for each observing segment absorbs the static part.

On this baseline, ocean loading contributes appreciably. We compute the solid-Earth tide following the IERS conventions[1] and the loading potential from FES2022,[24,25] including water attraction, induced Earth potential and clock displacement.[2] For the principal lunar semidiurnal tide, M_2, the differential solid-tide and loading amplitudes are 3.41×10^{-18} and 1.42×10^{-18} in fractional frequency, and their 20° phase difference gives a combined amplitude of 4.76×10^{-18} (Fig. 1c).

The observations span five campaigns and 54 observing segments over 353 days (Fig. 1b). The main analysis uses 545.9 h of retained samples to estimate template responses, spectra and signal-to-noise ratios. A filtered analysis of the same observations provides the time-domain displays (Fig. 2) and validation tests. Both analyses fit segment offsets; their selection, weights and template realizations are described in Methods and Extended Data Table 1a.

Amplitude, phase and coverage

The full-template response is 0.96 ± 0.11, where a response of one denotes agreement with the predicted change in frequency ratio (Fig. 3a). To separate the semidiurnal contributions, we group M_2 with the larger lunar elliptic semidiurnal tide, N_2, and the principal solar semidiurnal tide, S_2, with the lunisolar semidiurnal tide, K_2. Within each group, relative amplitudes and phases follow the physical model. The lunar-group response is 1.00 ± 0.13 (Fig. 3a), and allowing its phase to vary gives timing consistent with the prediction (Fig. 3b).

The lunar and solar semidiurnal periods are approximately 12.42 h and 12 h, and their changing relative phase allows the observations to distinguish them. The two-month summer campaign covers the full relative-phase cycle, whereas the shorter campaigns cover narrower intervals (Fig. 1d).

The five filtered-analysis campaign estimates are mutually consistent (Extended Data Fig. 1a,b and Extended Data Table 2a). As shown in Fig. 2, the summer time series and its dependence on lunar phase, after removal of the fitted offsets and non-lunar groups, agree well with the predicted lunar variation. Gaps are compressed on the time axis; Extended Data Fig. 2a shows the records of all campaigns, and Extended Data Fig. 2b,c shows the April campaign in calendar time.

The 274-h summer record was also used in the absolute frequency-ratio measurement of Hou et al.;[23] Extended Data Table 2b relates those estimates to the present analysis.

Separate semidiurnal detections

The equivalent-height amplitude spectrum (Fig. 3c) has lunar and solar-frequency peaks of 4.87 and 2.87 cm at 22.361 and 23.148 μHz. Subtracting the fixed unit-amplitude physical prediction reduces the amplitudes at these frequencies to 0.71 and 1.02 cm. Each principal frequency retains its peak when the other is controlled by a freely phased sinusoid, with N_2 and K_2 also controlled (Extended Data Fig. 3). A joint fit gives M_2 and S_2 amplitudes of 4.13 and 1.76 cm (4.50×10^{-18} and 1.92×10^{-18}; Extended Data Table 2c). These differ from the scan peaks because neighbouring frequencies are fitted simultaneously. The conditional spectra thus separate the M_2 and S_2-frequency contributions without imposing their astronomical phase relation. The predicted-phase group responses in Fig. 3a instead test agreement with the physical waveforms, keeping the relative phases of S_2 and K_2 fixed within their group (Methods).

Reliability of the separated signals

Reciprocal subtraction tests the coupling introduced by the sampling (Extended Data Fig. 4). Removing the jointly fitted lunar waveform changes the solar-frequency amplitude from 2.67 to 1.76 cm (2.918 to 1.921×10^{-18}). Removing the solar-frequency waveform changes the lunar amplitude from 4.45 to 4.13 cm (4.857 to 4.505×10^{-18}). Both remain identifiable. The maximum canonical correlation between the projected sine–cosine pairs is 0.290 (Extended Data Fig. 5), and the strongest sampling sidelobes respond to lunar-waveform subtraction as predicted (Extended Data Fig. 6).

The lunar estimate is also stable against specified daily disturbances. A 10^{-18} fractional-frequency sinusoid at 24 h or 12 h changes the main lunar response by at most 0.023 with a phase fixed across the record; with independent phases in each campaign, the bounds are 0.030 and 0.12, respectively. The filtered analysis provides complementary checks of daily terms, sampling and data selection (Fig. 4). Both clocks correct blackbody shifts in real time, and temperature regressions test residual correction errors (Methods and Extended Data Fig. 1d). In the filtered analysis, adding the fibre-link monitor as a regressor changes the lunar and full-template responses by less than 0.005. Retaining the screened transients leaves the lunar result consistent, and the fits improve predictions for most withheld observation periods (Methods). Extending the ocean mask near Shanghai by 0.3° or 0.6° changes the filtered lunar response by −0.05 or −0.08, respectively, about half of its statistical uncertainty.

Ocean loading and diurnal response

The separate ocean-loading and diurnal responses are constrained less precisely, with signal-to-noise ratios near unity (Table 1). The corresponding filtered-data fits are consistent with the prediction (Extended Data Fig. 1b; Supplementary Methods SM4 and SM5). Refined coastal loading models and longer continuous runs are expected to sharpen these comparisons.[26] These weaker constraints, compared with the lunar response, motivate a quantitative account of component precision.

Discussion

We develop a signal-to-noise model that relates component precision to signal strength, noise and phase coverage. The observation schedule enters through the sampling window and coupling between fitted components; within short segments, a slowly varying tide is partly absorbed by the fitted offset. For a specified estimator, the amplitude signal-to-noise ratio scales locally as

$$\rho_i(T) \simeq \rho_i(T_0)\frac{\eta_i(T)}{\eta_i(T_0)}\left(\frac{T}{T_0}\right)^{a_i-b_i}. \qquad (1)$$

Here T is the retained exposure and $\rho_i = S/R$. The exponents a_i and b_i describe how the phase-aligned signal and the noise standard deviation accumulate, and η_i accounts for the sampling window, weights and parameter coupling. For a coherent signal in white noise, $a_i = 1$ and $b_i = 1/2$, giving $\rho_i \propto \sqrt{T}$ at fixed η_i. We evaluate this white-noise case on the actual sampling with the measured amplitudes and a common clock-noise coefficient (Methods). Table 1 compares the resulting precision with the empirical errors of the same estimators.

Table 1 | Calculated and measured signal-to-noise ratios.

Evaluation	Lunar M_2	Solar S_2	Diurnal group	Ocean loading
S/R (white noise)	7.17	2.78	1.55	1.17
S/R (measured)	7.02	1.96	1.08	0.89

$S = |\hat{A}|$ is the measured amplitude or template-response magnitude and $R = u(\hat{A})$ its standard error. The white-noise row uses the common noise coefficient and the actual sampling; the measured row uses UTC-day-jackknife covariance. All entries use the auxiliary reduction: M_2 and S_2 from their joint sine–cosine fit, the diurnal group from the three-group fit and ocean loading from the two-template solid-Earth/loading fit (Methods).

As shown in Table 1, the lunar precision agrees well with the white-noise expectation, whereas the solar-frequency, diurnal and loading estimates have larger empirical uncertainties. Short segments constrain diurnal cycles weakly after offset removal, and the smaller loading signal overlaps the solid-Earth waveform. The signal-to-noise model thus accounts for most of the differences in precision between components. This comparison evaluates precision at the measured amplitudes; the exposure scaling in equation (1) remains conditional on the noise and sampling efficiency of future observations.

The model identifies distinct ways to improve future observations. Better clock stability lowers noise for a given retained exposure, whereas sustained network operation improves phase coverage and reduces component coupling. Continuous runs of at least two days with maintained phase lock cover successive diurnal cycles; a longer calendar span separates neighbouring frequencies, while gaps create sidelobes (Extended Data Fig. 6). Increasing retained exposure accumulates signal according to equation (1), with the applicable noise and sampling efficiency. Together with additional sites and local environmental records, these improvements would allow weaker potential changes to be identified after subtraction of the tidal background.

The measurement and separation of the lunar and solar-frequency semidiurnal contributions extend demonstrated sub-centimetre-equivalent clock performance[5] to time-varying geophysical signals. Together with the signal-to-noise model, they support the development of clock networks for geodesy, ultralight-dark-matter searches[27] and future clock-based gravitational-wave detectors.[28] Characterizing the tidal background has potential applications to the redefinition of the second[8] and the unification of height systems at the centimetre level.[9]

Note added. During the preparation of this manuscript, a related observation of tidal variations of the gravity potential in a European optical-clock network was reported;[29] the present work is complementary in separating the lunar and solar semidiurnal contributions on a coastal–inland baseline.

References

1. Petit, G. & Luzum, B. (eds) *IERS Conventions (2010).* IERS Technical Note **36** (2010). https://iers-conventions.obspm.fr/content/tn36.pdf
2. Farrell, W. E. Deformation of the Earth by surface loads. *Rev. Geophys.* **10**, 761–797 (1972). https://doi.org/10.1029/RG010i003p00761
3. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical clocks and relativity. *Science* **329**, 1630–1633 (2010). https://doi.org/10.1126/science.1192720
4. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nat. Commun.* **7**, 12443 (2016). https://doi.org/10.1038/ncomms12443
5. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). https://doi.org/10.1038/s41586-018-0738-2
6. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). https://doi.org/10.1088/1681-7575/ae449e
7. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *Metrologia* **63**, 065002 (2026). https://doi.org/10.1088/1681-7575/aeb809
8. Dimarcq, N. et al. Roadmap towards the redefinition of the second. *Metrologia* **61**, 012001 (2024). https://doi.org/10.1088/1681-7575/ad17d2
9. Wu, H., Müller, J. & Lämmerzahl, C. Clock networks for height system unification: a simulation study. *Geophys. J. Int.* **216**, 1594–1607 (2019). https://doi.org/10.1093/gji/ggy508

10. Arnold, K. J. et al. Lu^+ optical frequency references with accuracy verified at the 19th digit. *Nature* **658**, 60–64 (2026). https://doi.org/10.1038/s41586-026-11072-8

11. Bothwell, T. et al. Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). https://doi.org/10.1038/s41586-021-04349-7

12. Shen, Q. et al. Free-space dissemination of time and frequency with 10^{-19} instability over 113 km. *Nature* **610**, 661–666 (2022). https://doi.org/10.1038/s41586-022-05228-5

13. Caldwell, E. D. et al. Quantum-limited optical time transfer for future geosynchronous links. *Nature* **618**, 721–726 (2023). https://doi.org/10.1038/s41586-023-06032-5

14. Gozzard, D. Clocks synchronized at the quantum limit. *Nature* **618**, 680–681 (2023). https://doi.org/10.1038/d41586-023-01937-7

15. Kómár, P. et al. A quantum network of clocks. *Nat. Phys.* **10**, 582–587 (2014). https://doi.org/10.1038/nphys3000

16. Nichol, B. C. et al. An elementary quantum network of entangled optical atomic clocks. *Nature* **609**, 689–694 (2022). https://doi.org/10.1038/s41586-022-05088-z

17. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nat. Phys.* **14**, 437–441 (2018). https://doi.org/10.1038/s41567-017-0042-3

18. Tapley, B. D., Bettadpur, S., Ries, J. C., Thompson, P. F. & Watkins, M. M. GRACE measurements of mass variability in the Earth system. *Science* **305**, 503–505 (2004). https://doi.org/10.1126/science.1099192

19. Rymer, H. & Brown, G. Gravity changes as a precursor to volcanic eruption at Poás volcano, Costa Rica. *Nature* **342**, 902–905 (1989). https://doi.org/10.1038/342902a0

20. Lynett, P. et al. Diverse tsunamigenesis triggered by the Hunga Tonga-Hunga Ha’apai eruption. *Nature* **609**, 728–733 (2022). https://doi.org/10.1038/s41586-022-05170-6

21. Rodell, M. et al. Emerging trends in global freshwater availability. *Nature* **557**, 651–659 (2018). https://doi.org/10.1038/s41586-018-0123-1

22. Chen, F.-X. et al. 10^{-21}-Level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light Sci. Appl.* **15**, 276 (2026). https://doi.org/10.1038/s41377-026-02299-1

23. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. Preprint at https://arxiv.org/abs/2610.34250v1 (2026).

24. Lyard, F. H., Allain, D. J., Cancet, M., Carrère, L. & Picot, N. FES2014 global ocean tide atlas: design and performance. *Ocean Sci.* **17**, 615–649 (2021). https://doi.org/10.5194/os-17-615-2021

25. CNES. *FES2022 (Finite Element Solution) Tidal model (Version 2024)* [Data set] (2024). https://doi.org/10.24400/527896/A01-2024.004

26. Schröder, S., Stellmer, S. & Kusche, J. Potential and scientific requirements of optical clock networks for validating satellite-derived time-variable gravity data. *Geophys. J. Int.* **226**, 764–779 (2021). https://doi.org/10.1093/gji/ggab132

27. Boulder Atomic Clock Optical Network (BACON) Collaboration. Frequency ratio measurements at 18-digit accuracy using an optical clock network. *Nature* **591**, 564–569 (2021). https://doi.org/10.1038/s41586-021-03253-4

28. Kolkowitz, S. et al. Gravitational wave detection with optical lattice atomic clocks. *Phys. Rev. D* **94**, 124043 (2016). https://doi.org/10.1103/PhysRevD.94.124043

29. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. Preprint at https://arxiv.org/abs/2610.39635v1 (2026).

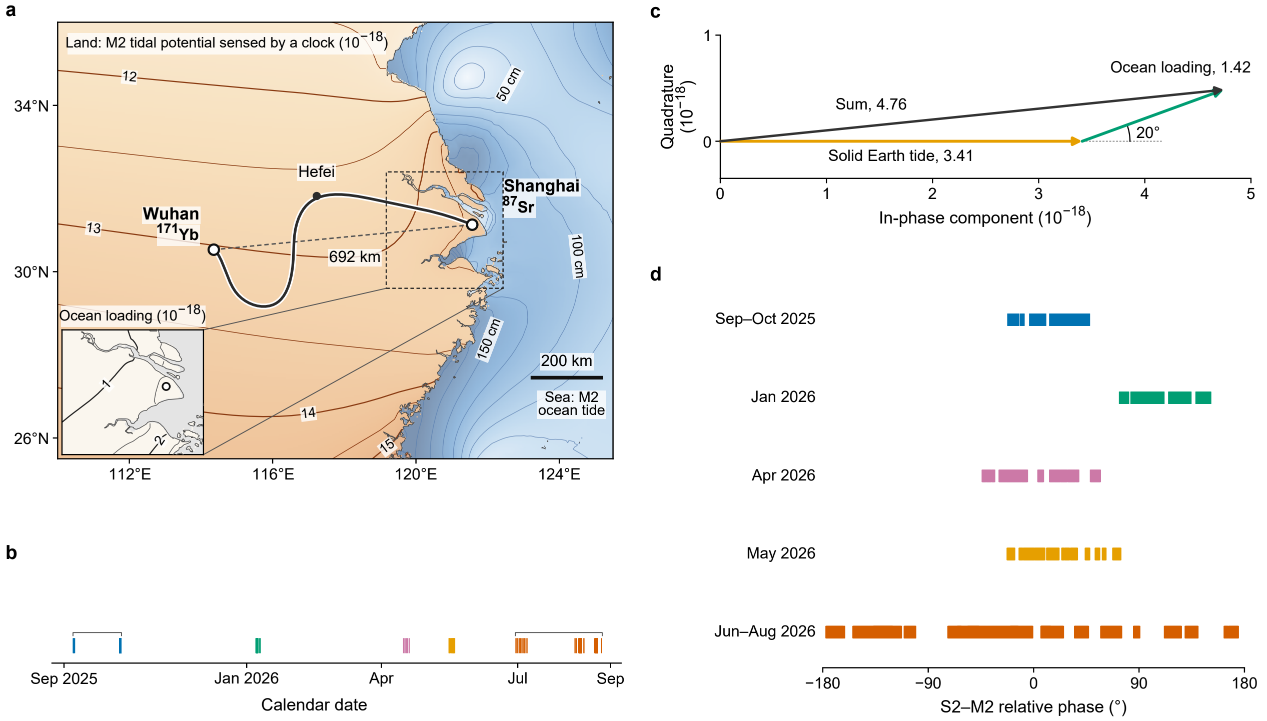


Fig. 1 | The predicted tide on a coastal–inland baseline. **a,** The Sr clock in Shanghai, the Yb clock in Wuhan and the fibre link through Hefei; the dashed line marks the 692-km distance between the clocks. Land contours show the predicted M_2 tidal potential sensed by a clock, including the solid Earth tide and ocean loading, in units of 10^{-18} in fractional frequency (interval 0.5; integer contours labelled). Sea shading and contours show the FES2022 M_2 ocean-tide amplitude (interval 25 cm; 50, 100 and 150 cm labelled). The inset shows the ocean-loading contribution within the dashed box around Shanghai, with labelled contours at 1 and 2×10^{-18}. **b,** Observation calendar of the five campaigns from September 2025 to August 2026. Bars mark the retained observation intervals in the filtered validation analysis; brackets span campaigns with separated runs. **c,** Predicted differential M_2 phasors of the solid Earth tide, the ocean-loading potential and their sum, referenced to the solid Earth tide; the loading phasor is at 20° to the solid Earth tide. **d,** Relative astronomical phase of S_2 and M_2, including the nodal correction, at the filtered 600-s samples, shown as one tick per sample on a −180° to 180° axis. The short campaigns cover limited ranges of relative phase, whereas the summer campaign covers the full cycle, with some gaps, as the lunar tide moves through the solar day.

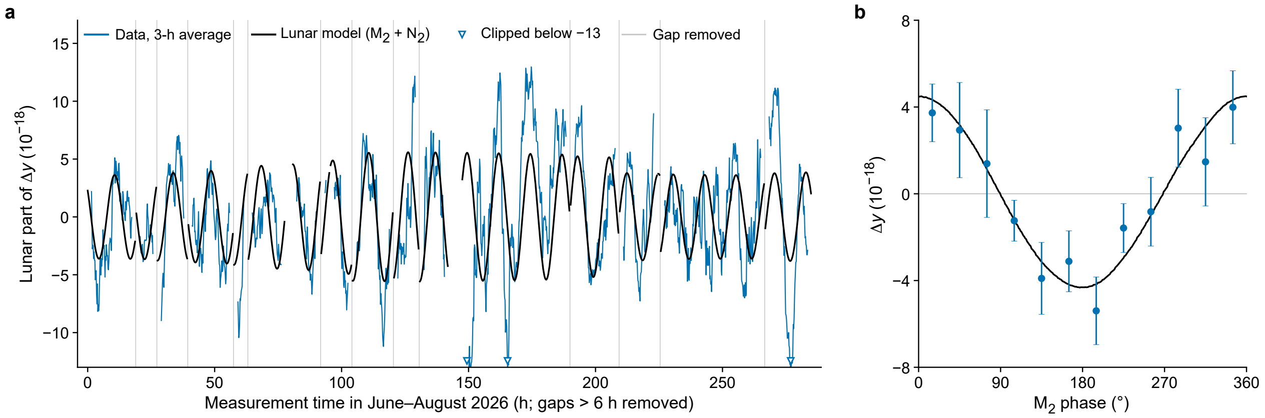
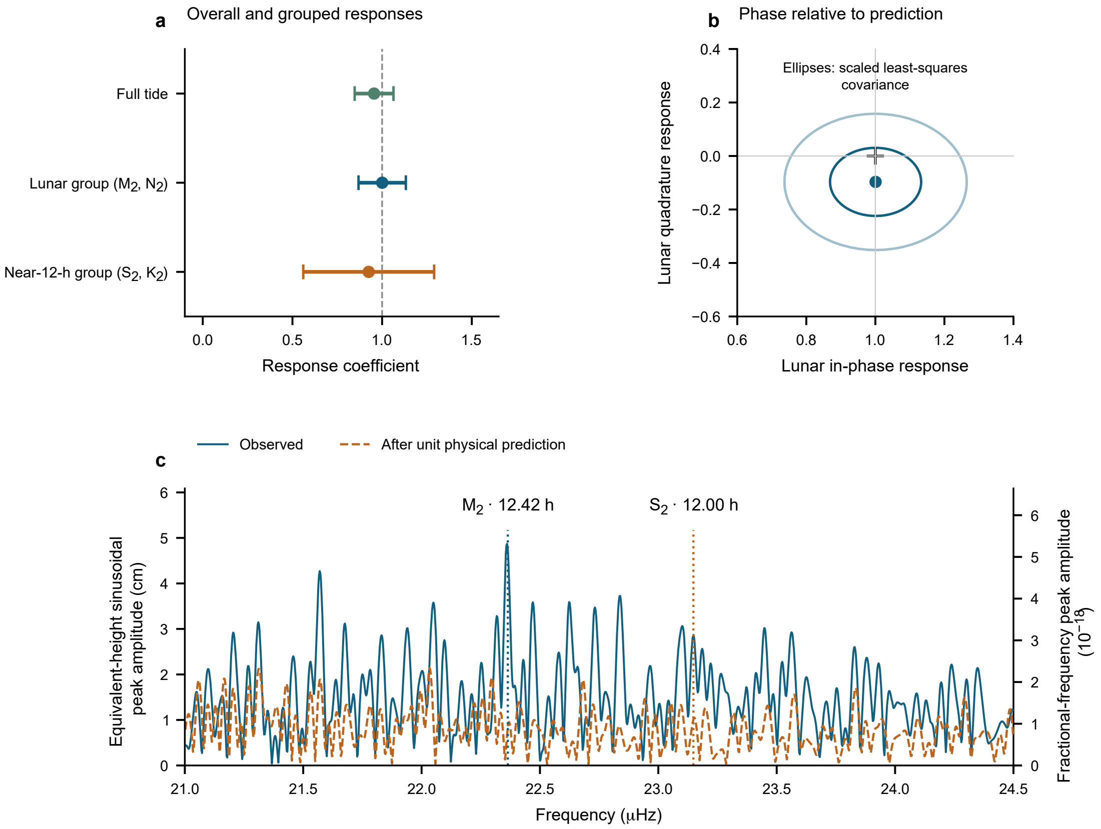


Fig. 2 | The lunar tide in the summer record. Both panels use the filtered validation analysis (Methods). **a,** Three-hour running averages of the June–August 2026 frequency ratio after subtracting fitted segment offsets and the fitted near-12-h and diurnal (non-lunar) tidal groups (blue), with the lunar model (black). The measurement-time axis compresses gaps longer than 6 h; grey lines mark their removal, and values below -13×10^{-18} are clipped (open triangles). White frequency noise alone corresponds to a scatter of approximately 5×10^{-18} for a 3-h average. **b,** The same data folded at the model M_2 phase, relative to the predicted maximum, in 30° bins, with the lunar model (M_2 and N_2) folded on identical sampling. Error bars are day-by-day jackknife uncertainties with the fitted non-lunar terms held fixed. No empirical daily terms are fitted.



Rayleigh scale 1/Tspan = 0.03275 μHz; base grid step = 0.002729 μHz

Fig. 3 | Amplitude, phase and frequency of the tidal response. All panels use the main analysis (the auxiliary reduction) into 3,299 non-overlapping 600-s means with 54 segment constants. **a,** Complete-template ('full tide') response and jointly fitted lunar (M_2, N_2) and near-12-h (S_2, K_2) group responses, from separate fits with period-dependent noise weights. Error bars are one standard error, the larger of the scaled least-squares and UTC-day-deletion jackknife errors. **b,** Lunar-group in-phase and quadrature coefficients; the cross marks $(1, 0)$, and contours have radii one and two in the scaled least-squares covariance. The quoted marginal errors, the larger of the scaled least-squares and UTC-day-jackknife values, are 0.13 and 0.20; the scaled least-squares quadrature error is 0.127. **c,** Equivalent-height sinusoidal peak-amplitude spectra over 21–24.5 μHz, before and after subtracting the fixed unit-amplitude physical prediction, with equal weights and segment constants projected from data and trial sinusoids. The left and right axes give equivalent height and fractional-frequency amplitude, related by $A_{h,\mathrm{cm}} = 100c^2A_y/g_0$ with $g_0 = 9.80665\ \mathrm{m/s^2}$; both are linear. Markers indicate the M_2 and S_2 frequencies; the nominal Rayleigh scale (0.03275 μHz) and grid step (0.002729 μHz) are annotated.

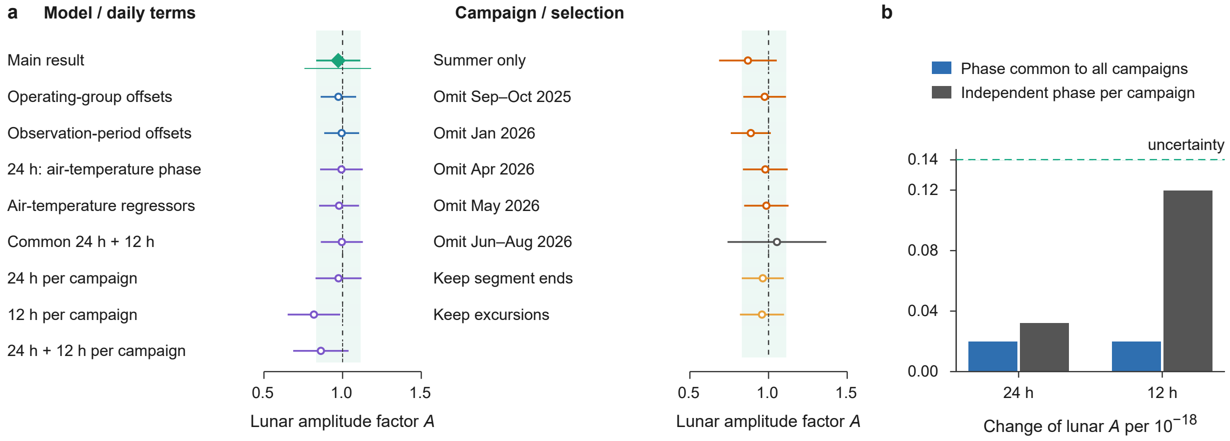


Fig. 4 | Robustness of the lunar amplitude. Panel a spans the two left columns, which share an amplitude range of 0.5–1.5; panel b is on the right. Both panels use the filtered validation analysis (the primary reduction). **a,** Lunar amplitude for the primary fit (filled diamond and shaded uncertainty band), alternative offset definitions, daily terms represented by air-temperature measurements or fitted 24-h and 12-h sinusoids, the summer campaign alone, omission of each campaign, and retention of segment ends or transient excursions. Error bars and the shaded band indicate one standard deviation. The thin bar below the primary result is the 95% interval from resampling operating groups. These estimates share observations. The grey interval is obtained without the summer campaign. **b,** Change in the lunar amplitude caused by a 24-h or 12-h sinusoid of fractional-frequency amplitude 10^{-18}, with one phase common to all campaigns or with the combination of independent campaign phases that produces the largest change. The dashed line marks the uncertainty of the primary estimate.

Methods

Clocks, link and data

The Sr optical lattice clock is in Shanghai (31.130° N, 121.570° E, height approximately 5 m) and the Yb optical lattice clock in Wuhan (30.529° N, 114.357° E, approximately 30 m); changing the Shanghai height by 5 m changes the differential tidal model by only 4×10^{-23}. The static geopotential difference, 280.042 ± 0.085 m^2 s^{-2} from first-order levelling [23], enters the frequency ratio as a constant absorbed by the fitted offsets. The evaluated systematic uncertainties of the Sr and Yb clocks are 9.2×10^{-19} and 1.3×10^{-18} [6,7]; the clocks, fibre link and frequency chain are described in ref. 23, and optical frequency dissemination [22,30] over field fibres of comparable length has been demonstrated.

A 1,550-nm carrier referenced to the Sr clock through a frequency comb in Shanghai is transmitted to Wuhan, where its beat against the Yb-referenced comb is counted at 1-s intervals. In 2026 the beat frequency is 33.6 MHz, and a deviation of 1 Hz corresponds to 5.17×10^{-15} in Yb/Sr; in 2025 the counter recorded the beat after division by two, which is accounted for. The white-frequency-noise level of the tide-corrected ratio, $\sigma_y\sqrt{\tau}$, from the overlapping Allan deviation [31] at 256–4,096 s, is 7.9×10^{-16} in September–October 2025 and 4.2–5.3×10^{-16} in the 2026 campaigns, evaluated for each campaign in the primary reduction.

The 1,350-km link extends a 2,067-km dissemination system with bias-free noise compensation based on digital radio-frequency phase recording [22,23]. The out-of-loop round-trip signal and the optical phase-locked loops of the frequency chain are monitored continuously [23]; maintaining lock preserves the phase connection needed for coherent accumulation.

The observations cover five campaigns (September–October 2025, January 2026, April 2026, May 2026 and June–August 2026) and 54 segments defined by the experiment team; the summer segments are those of ref. 23. The overall effective observing time is 545.9 h, summed over the retained one-second samples in the main analysis. There are 38 segments before June and 16 in June–August. Campaigns define the seasonal subsets and campaign-specific daily terms, the eight observation periods are the units of held-out prediction, the 29 operating groups are the units of group-bootstrap resampling, and segments carry the fitted offsets. Time stamps are in local time (UTC+8).

Main analysis: grouped responses

The main analysis (labelled the auxiliary reduction in the supplementary material) provides the summary responses, spectra and signal-to-noise ratios. The filtered validation analysis (labelled the primary reduction in the figure and supplementary labels) provides the time-domain displays and robustness checks. These analyses share observations and are not independent measurements (Extended Data Table 1a).

Figure 3 and Extended Data Figs. 3–6 use $N_\mathrm{b} = 3,299$ non-overlapping 600-s means over a calendar span of 353.39 days and 54 segment constants. Within each predefined observing segment, beat-frequency records differing by more than 10 Hz from the segment median were excluded. Retained records were averaged separately within each segment on a 600-s wall-clock grid in Beijing time (UTC+8), retaining bins with at least 180 nominal one-second members. Bin timestamps denote the nominal grid centres. This reduction applies neither the triangular filter nor the additional running-median transient screening used in the filtered validation analysis. Its solid-Earth template converts the IERS radial displacements into fractional frequency,

$$y_\mathrm{solid} = -\frac{1 + k_2 - h_2}{h_2} \frac{g(u_\mathrm{Wuhan} - u_\mathrm{Shanghai})}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794$ m/s^2, which differ from the IERS values of the filtered-validation model ($k_{22} = 0.29801$) by 0.4% in $1 + k_2 - h_2$; its loading template uses the effective-potential convolution with 18 constituents. The lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal (K_1, O_1, P_1, Q_1) groups comprise eight constituents; the remaining terms form a fixed remainder r_0. The complete-template and grouped models are $y_i = c_{s(i)} + A_T(s_{L,i} + s_{S,i} + s_{D,i} + r_{0,i}) + \epsilon_i$ and $y_i - r_{0,i} = c_{s(i)} + A_L s_{L,i} + A_S s_{S,i} + A_D s_{D,i} + \epsilon_i$. The phase-relaxed model adds a quarter-cycle-shifted lunar template while keeping the M_2/N_2 relation.

The weights are $W_{ii} = 1/\sigma_i^2$ with $\sigma_i = \alpha_{p(i)}/\sqrt{\Delta t_i}$, where $\Delta t_i = n_i \delta t$ is the retained exposure of bin i (n_i one-second members, $\delta t = 1$ s) and α_p is the white-frequency-noise coefficient of observing period p from overlapping Allan deviations at 256–4,096 s: 9.70 and 5.70×10^{-16} for September and October 2025, and 4.30–5.76×10^{-16} for the six 2026 periods. Errors are the larger of the residual-scaled least-squares and UTC-day-deletion jackknife errors, each deletion refitting all coefficients. The contours in Fig. 3b show the scaled least-squares covariance at radii one and two.

Spectra and constituent separation

For Fig. 3c, at each trial frequency we fit a freely phased sinusoid and the 54 segment constants to the auxiliary means with equal weights,

$$y_i = c_{s(i)} + a_f \cos[2\pi f(t_i - t_0)] + b_f \sin[2\pi f(t_i - t_0)] + \epsilon_i,$$

and plot the peak amplitude $A_y(f) = \sqrt{a_f^2 + b_f^2}$; for the residual curve, the fixed unit-amplitude physical prediction is subtracted first. The equivalent height is $A_{h,\text{cm}} = 100c^2 A_y/g_0$ with $g_0 = 9.80665\ \text{m/s}^2$. The span $T_{\text{span}} = 353.3889$ d sets the nominal Rayleigh scale $1/T_{\text{span}} = 0.03275$ μHz. The scan spacing is $1/(12T_{\text{span}}) = 0.002729$ μHz, with the astronomical frequencies added, and the peaks are local maxima within one Rayleigh scale of each prescribed frequency.

For Extended Data Figs. 3–6, a nuisance matrix N (segment constants, and free N_2 and K_2 pairs where stated) is projected out of the data and of each trial cosine–sine pair X_f with $R_N = I - QQ^\top$, where Q is an orthonormal basis of N. The fitted coefficients $\hat{\boldsymbol{\beta}}_f = (R_N X_f)^+ R_N y$, with y in units of 10^{-18}, give the equivalent-height peak amplitude $A_{h,\text{cm}}(f) = (c^2/g_0)\,10^{-16}\,\|\hat{\boldsymbol{\beta}}_f\|$, as in Fig. 3c. At a frequency coinciding with a controlled constituent, the trial pair lies in the nuisance space and its amplitude is undefined; these entries are not plotted. Phases refer to 6 September 2025, 16:05 UTC. The M_2 and S_2 waveforms come from a simultaneous four-coefficient fit with no amplitude or phase constraint between them; without N_2/K_2 control their amplitudes are 4.63 and 1.82×10^{-18}. Conditional scans add the other principal pair to N and scan within ±0.16 μHz of the target. Compensation tests subtract a scaled donor waveform and re-estimate the target with the same nuisance terms, and canonical correlations are the singular values of $Q_M^\top Q_S$ for the residualized pairs (Supplementary Methods SM7).

Signal-to-noise ratios

We define $\rho_i = |\hat{A}_i|/u(\hat{A}_i)$. For the white-noise case of equation (1), $H = (X^\mathsf{T} W X)^{-1} X^\mathsf{T} W$ maps the observations to the fitted coefficients for design matrix X and weights W. For white frequency noise with $\alpha = 5.4 \times 10^{-16}\sqrt{\text{s}}$, the bin variance is $\alpha^2/\Delta t_k$, giving

$$u_{i,\text{white}}^2 = \alpha^2 \sum_k \frac{H_{ik}^2}{\Delta t_k}, \qquad \rho_{i,\text{white}} = \frac{|\hat{A}_i|}{u_{i,\text{white}}}. \tag{2}$$

With the retained exposure $T = \sum_{k=1}^{N_\text{b}} \Delta t_k$ of the N_b means, $s_{i,\text{rms}}^2 = \sum_k \Delta t_k \, s_i(t_k)^2/T$ for template s_i per unit response and $\eta_i = \alpha/(s_{i,\text{rms}}\sqrt{T}\, u_{i,\text{white}})$,

$$\rho_{i,\text{white}} = \frac{|\hat{A}_i| s_{i,\text{rms}}}{\alpha}\, \eta_i \sqrt{T}. \tag{3}$$

The sampling window, offsets, template coupling and weights enter η_i, and equation (3) gives the $a_i = 1$, $b_i = 1/2$ sensitivity at the measured amplitude (Extended Data Table 2d). For M_2 and S_2 in Table 1, $\hat{A} = (\beta_c^2 + \beta_s^2)^{1/2}$ from the equal-weight joint fit of M_2, S_2, N_2 and K_2 pairs with 54 segment offsets, and $u^2(\hat{A}) = J C_i J^\mathsf{T}$ with $J = (\beta_c, \beta_s)/\hat{A}$; $C_\beta = H \Sigma H^\mathsf{T}$ with $\Sigma_{kk} = \alpha^2/\Delta t_k$ for the white-noise row, and the covariance of 43 UTC-day deletion fits for the measured row. The diurnal column uses the three-group fit. The loading column uses a fit of the complete solid-Earth and loading templates with two free coefficients, 54 segment offsets and equal weights, with UTC-day-deletion errors (Extended Data Table 2d and Supplementary Methods SM6).

Filtered validation: data selection

Every segment is processed identically. The time axis is reconstructed from the first time stamp and the sample index. Samples more than 10 Hz (approximately 5×10^{-14} in Yb/Sr) from the segment median are removed, the longest uninterrupted run is retained, and leading and trailing samples deviating from the median by more than 1% of the peak-to-peak range are trimmed. To remove smaller transient excursions, 300-s averages are compared with a running median over ±30 min, and averages deviating by more than six robust standard deviations are removed together with their neighbours. This step removes 0.4% of the samples, or 2.2 h (Extended Data Fig. 7d–f). The criterion uses no tidal model, and adding the model to the data leaves the selection unchanged; because the subsequent filter is linear, injection tests use the selected data. Screening variants are given in Supplementary Methods SM2.

Filtered validation: tidal model

The solid Earth contribution at each clock is

$$\delta W = \sum_m (1 + k_{2m}) V_{2m} + (1 + k_3) V_3 - g u_r,$$

where V_{2m} are the degree-2 tide-generating potentials of the Moon and Sun of order m, V_3 is the lunar degree-3 potential and u_r is the radial displacement. Love numbers and displacements follow the IERS 2010 conventions [1] (Supplementary Methods SM1). The ocean-loading potential is computed by grid convolution,

$$\delta W_{\text{ocean}}(t) = \sum_{\text{ocean}} \rho_w \zeta(\mathbf{r}', t) G_W(\psi)\, \mathrm{d}A, \qquad G_W(\psi) = \frac{ga}{M_E} \sum_n (1 + k'_n - h'_n) P_n(\cos\psi),$$

with the FES2022 ocean elevation ζ on a $1/30°$ grid [24,25], $\rho_w = 1,025$ kg m^{-3}, Earth's radius a and mass M_E, and angular distance ψ. The Green's function [2] uses PREM load Love numbers [32,33]. After removal of the ocean mean, eight constituents (M_2, S_2, N_2, K_2 and the diurnal K_1, O_1, P_1, Q_1) are synthesized with astronomical arguments and nodal corrections. They explain 99.84% of the model variance; the remainder, with an r.m.s. of 0.18×10^{-18}, is held at its predicted value. The radial displacement from the same convolution agrees with the FES2022 load-tide grid to within 2.2% in amplitude and 1.2° in phase for seven of the eight constituents; for S_2 the grid derives from a different ocean-model release, and substituting it changes the full-template response by 0.001 (Extended Data Fig. 8; Supplementary Methods SM1). Extending the ocean mask near Shanghai by 0.3° or 0.6° changes the lunar response by −0.05 or −0.08, respectively, about half of its statistical uncertainty of 0.14.

Filtered validation: uncertainties

The 1-s observations and model are filtered identically with a 1,200-s triangular window stepped by 600 s and fitted by least squares as $y(t) = c_i + As(t)$, with an offset c_i for segment i. For band analyses, the eight-constituent model is divided into semidiurnal (M_2, S_2, N_2, K_2) and diurnal (K_1, O_1, P_1, Q_1) bands, or into lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal groups with separate amplitudes. The primary grouped fit contains no empirical daily terms. Separate fits of the solid-Earth and loading templates assess their coupling (Supplementary Methods SM4).

For the full template, the semidiurnal band and the lunar group, uncertainties are calibrated by residual resampling. The fitted residuals are shifted circularly within each continuous run of the filtered record, added to the model with $A = 1$ and refitted; the standard deviation over 200 realizations gives the error, retaining the within-run residual structure. For the diurnal and near-12-h groups, fits with empirical daily terms and fitted daily amplitudes, we quote the larger of the scaled least-squares standard error[34] and the day-by-day jackknife error; in the resampled data these differ from the calibrated uncertainty by factors of 0.8–1.3. Group-bootstrap intervals draw the 29 operating groups with replacement in 1,000 refits; the segment bootstrap draws the 54 segments. In the phase test, each band has in-phase and quadrature templates, the latter shifting every constituent by 90°, with segment-bootstrap uncertainties. Quoted uncertainties are one standard deviation (Supplementary Methods SM3).

The phase-folded summer display (Fig. 2b) subtracts the fitted offsets and non-lunar groups and averages in 30° M_2-phase bins; the M_2+N_2 model is folded with the same sampling, and the day-by-day jackknife error bars hold the fitted non-lunar terms fixed. The time-series display uses 3-h averages, with gaps longer than 6 h removed from the horizontal axis and marked.

The full-record and April spectra (Extended Data Fig. 7c and Extended Data Fig. 2c) show sinusoidal power, the squared amplitude divided by two, after projection of segment offsets and before and after subtraction of the fitted whole template. Their thresholds are 99% quantiles of the maximum over 5–40 μHz from 200 resampled residual series (full record) and over 15–50 μHz from 1,000 series (April, together with pointwise 99% levels). No April frequency exceeds the latter threshold after tidal correction.

Daily terms, environment and prediction

In the filtered validation analysis, daily sensitivity fits add $a_1 \cos \omega t + b_1 \sin \omega t + a_2 \cos 2\omega t + b_2 \sin 2\omega t$, where $\omega = 2\pi/(1 \text{ day})$ and t is local time, with common or campaign-specific coefficients. Alternatives use, in each campaign, a free-amplitude 24-h term with the phase of the 24-h harmonic of hourly ERA5 2-m air temperature at the two sites [35], or the air-temperature anomalies at both sites relative to a 24-h running mean as two regressors.

The response of the lunar estimate to a disturbance $B \cos(\omega t - \varphi)$ follows from the lunar row h of the estimator H (defined in the signal-to-noise calculation) for the specified fit. For a common phase, the maximum amplitude change is

$$B\sqrt{(h \cdot \cos \omega t)^2 + (h \cdot \sin \omega t)^2};$$

for phases independent between campaigns, it is the sum of the corresponding campaign norms. The main-analysis bounds reported in Results use the three-group design, with the main sample times, lunar, near-12-h and diurnal templates, segment offsets and observing-period-noise weights defined in Methods. Figure 4 uses the corresponding operator for the equal-weight filtered validation fit. The following injection, environmental and held-out prediction checks use the filtered record.

Injection tests add either the full tidal model ($A = 1$) or the model without its lunar group ($A = 0$) to circularly shifted residuals of the full-template fit, under four daily-disturbance scenarios: none; campaign-specific 24-h terms with air-temperature phases and fitted amplitudes; common 24-h and 12-h terms with fitted amplitudes and phases; and the fitted summer 24-h and 12-h terms. With 300 realizations per scenario and injected amplitude, the lunar-amplitude biases lie between −0.03 and 0.00, and for $A = 1$ the fitted amplitude exceeds three standard errors in every realization (Extended Data Table 1b).

Campaign information shares project the lunar template onto the complement of the near-12-h and diurnal templates and segment offsets, and sum the squared projection within each campaign. Held-out prediction fits the amplitude to all but one observation period and applies it to the excluded period, refitting only that period's offsets (Supplementary Methods SM4).

The round-trip beat of the fibre link, recorded in Shanghai near 10 MHz at 1-s resolution, is normalized by the optical carrier frequency and filtered like the clock ratio. It covers 474 h of the analysed data. Including it as an additional regressor, or excluding 300-s blocks in which it deviates by more than 5×10^{-16}, changes the lunar and whole-template responses by less than 0.005.

The Sr vacuum-chamber temperature, the mean of 20 PT100 sensors sampled every 20 s, is available from 28 June to 7 July 2026 and overlaps 98 h of the filtered summer record. The blackbody sensitivity is 6.26×10^{-17} K^{-1} at 294.3 K. Because both clocks correct blackbody shifts in real time, a regression of the ratio on the chamber-temperature shift tests residual correction errors and has an expected coefficient of zero; it gives 0.92 ± 0.74 (Supplementary Methods SM4). ### Use of large language models

Large language models were used to assist with code writing, manuscript drafting and editing. All scientific content, analyses and conclusions are those of the authors, who independently verified all numerical results and take full responsibility for the manuscript.

Data availability

The frequency-ratio records, the segment table with the retained samples and the tidal-model time series supporting the findings of this study are available from the corresponding author upon reasonable request.

Code availability

The analysis code for the primary and auxiliary reductions is available from the corresponding author upon reasonable request.

Methods references

30. Schioppo, M. et al. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through a 2220 km optical fibre network. *Nat. Commun.* **13**, 212 (2022). https://doi.org/10.1038/s41467-021-27884-3

31. Riley, W. J. *Handbook of Frequency Stability Analysis.* NIST Special Publication **1065** (2008). https://doi.org/10.6028/NIST.SP.1065

32. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophys. J. Int.* **120**, 287–311 (1995). https://doi.org/10.1111/j.1365-246x.1995.tb01819.x

33. Sutterley, T., Velicogna, I., Mohajerani, Y. & Lecomte, H. *gravity-toolkit*, version 1.2.8. Zenodo https://doi.org/10.5281/zenodo.23107010 (2026).

34. Birge, R. T. The calculation of errors by the method of least squares. *Phys. Rev.* **40**, 207–227 (1932). https://doi.org/10.1103/PhysRev.40.207

35. Hersbach, H. et al. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* **146**, 1999–2049 (2020). https://doi.org/10.1002/qj.3803

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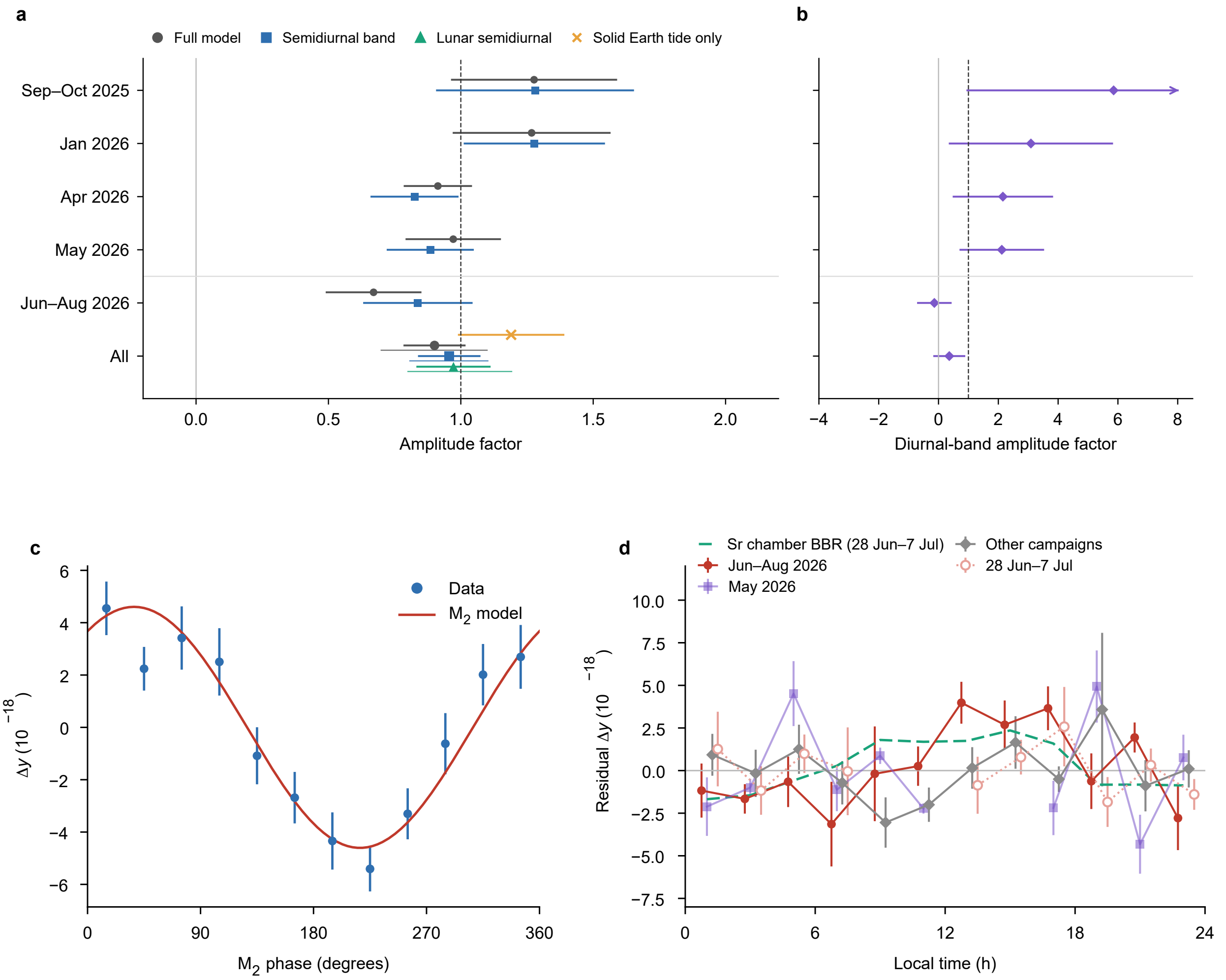
Competing interests

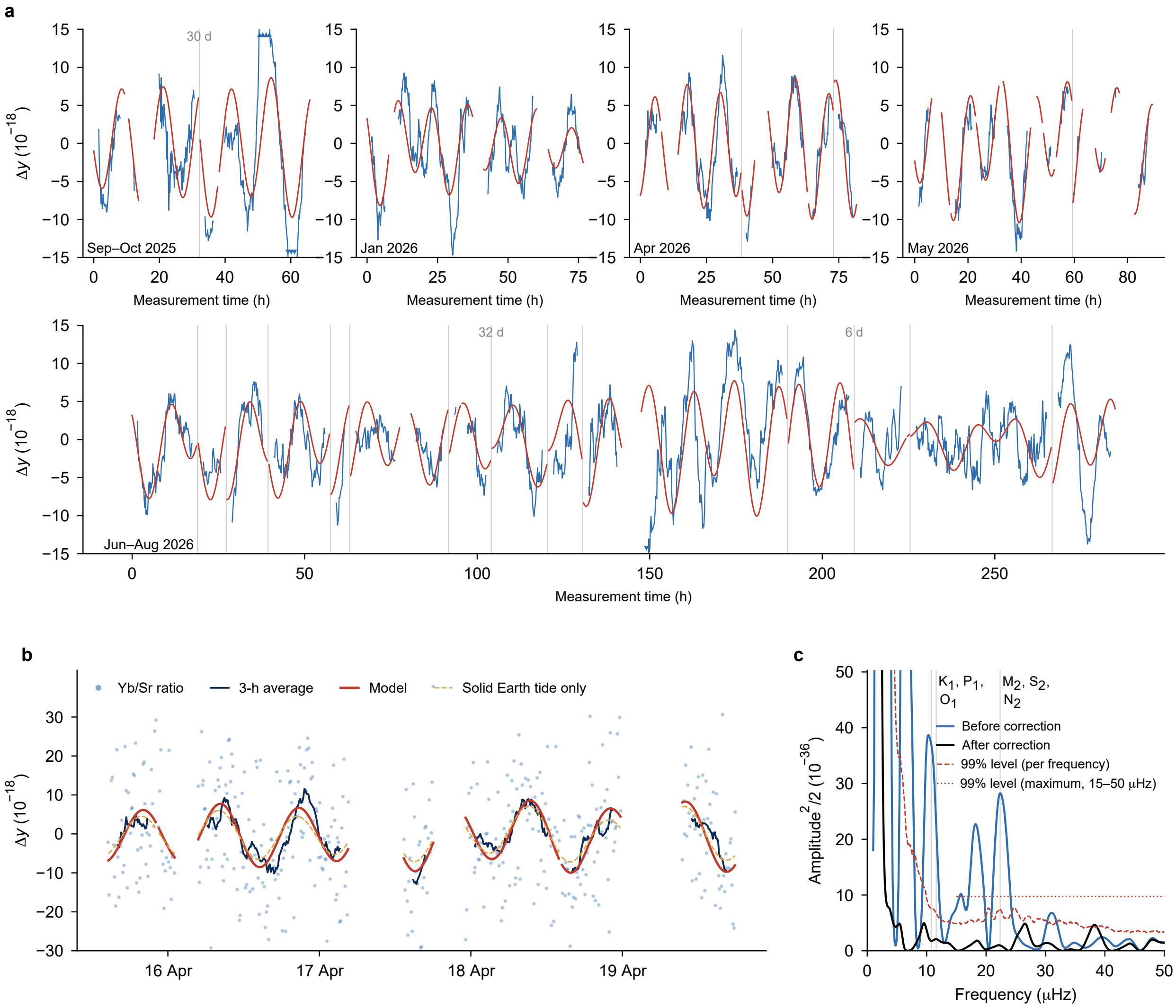
The authors declare no competing interests.

Additional information

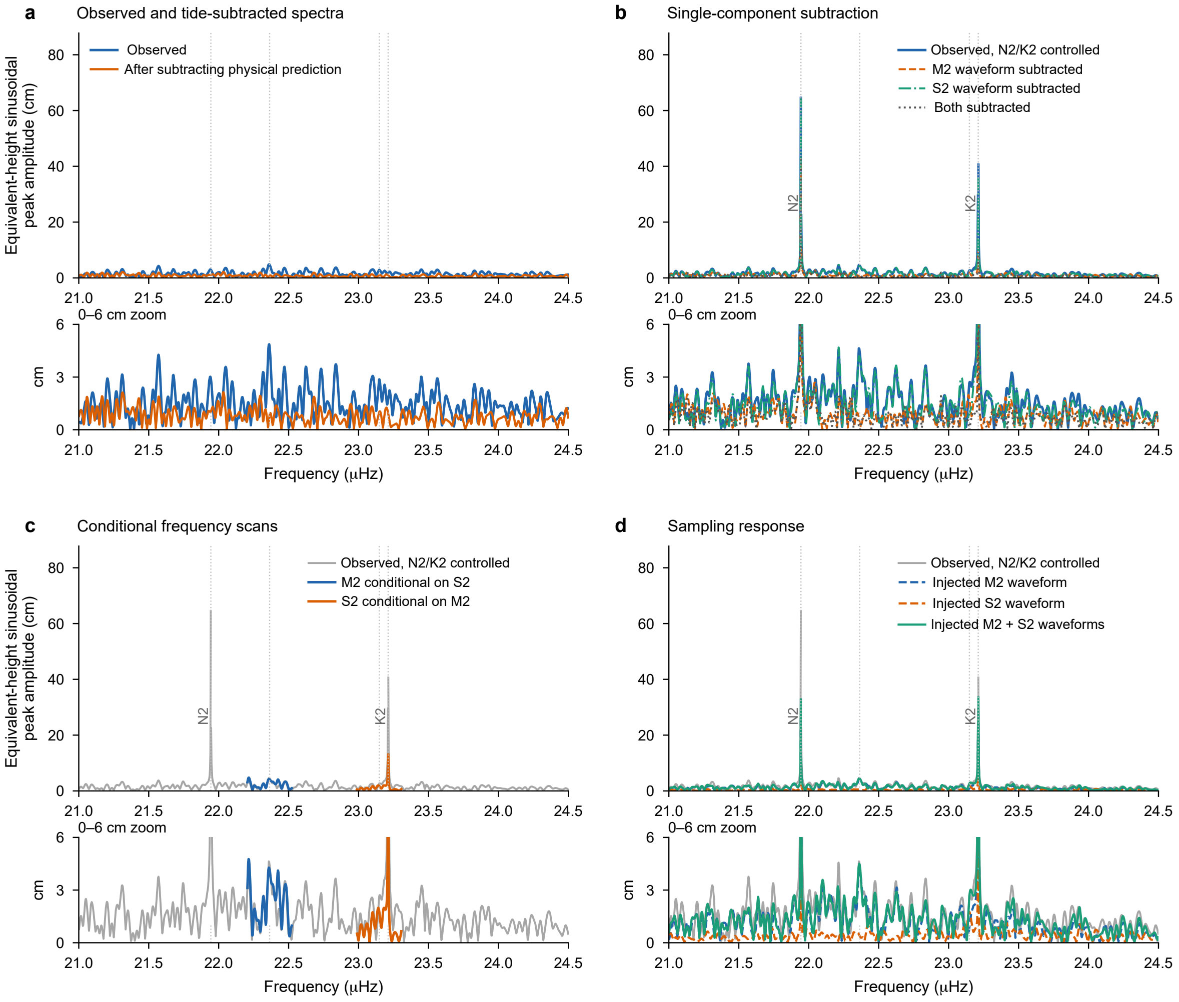
Supplementary information is available for this paper.



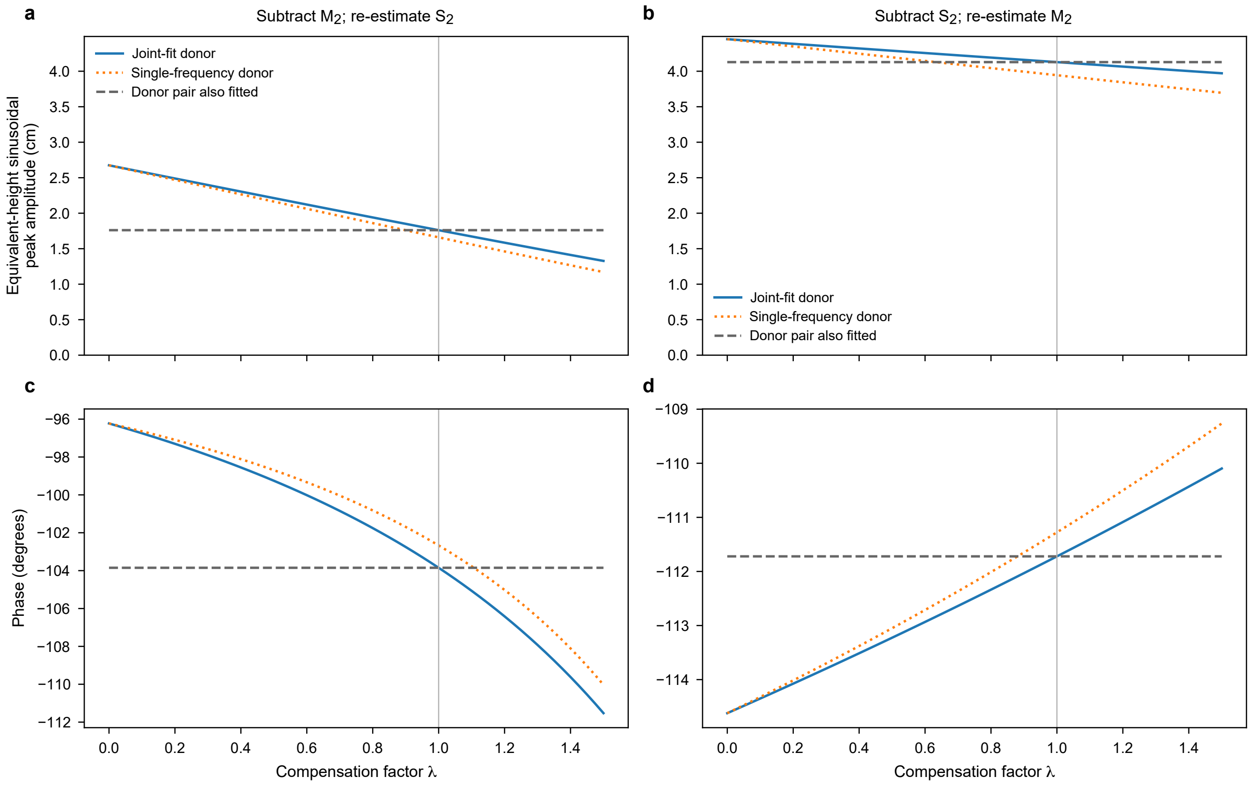
Extended Data Fig. 1 | Campaign-resolved responses and phase folds. **a,** Full-model and semidiurnal amplitude factors for each campaign and the combined observations, fitted with segment offsets. Combined-data estimates also include the lunar group and the solid Earth tide alone. **b,** Diurnal amplitude fitted jointly with the semidiurnal band in a two-band fit; the combined estimate is 0.36 ± 0.50. The arrow marks an uncertainty extending beyond the plotted range. Error bars represent one standard deviation, calibrated by residual resampling for the full model, semidiurnal band and lunar group; for the diurnal band and solid Earth tide alone they are the larger of the scaled least-squares and day-by-day jackknife uncertainties. Thin lines in the combined-data rows show 95% intervals from resampling operating groups, or segments for the lunar group. **c,** All observations folded at the astronomical M_2 phase after subtracting fitted segment offsets and all tidal constituents except M_2 from the full-model fit. Points are means in 30° bins with day-by-day jackknife uncertainties, with the removed terms held fixed; the curve is the M_2 model. **d,** Tidally corrected residuals folded by local time in 2-h bins for summer, May (light symbols), the other three campaigns, and summer observations overlapping the temperature record from 28 June to 7 July (open circles). Error bars are day-by-day jackknife uncertainties; bins covering fewer than three days are omitted. The dashed curve is the mean daily blackbody-radiation (BBR) shift computed from the Sr chamber temperature over the same interval, after removing a 24-h running mean and excluding the temperature transient on 30 June; the real-time correction removes this shift from the recorded frequency ratio.



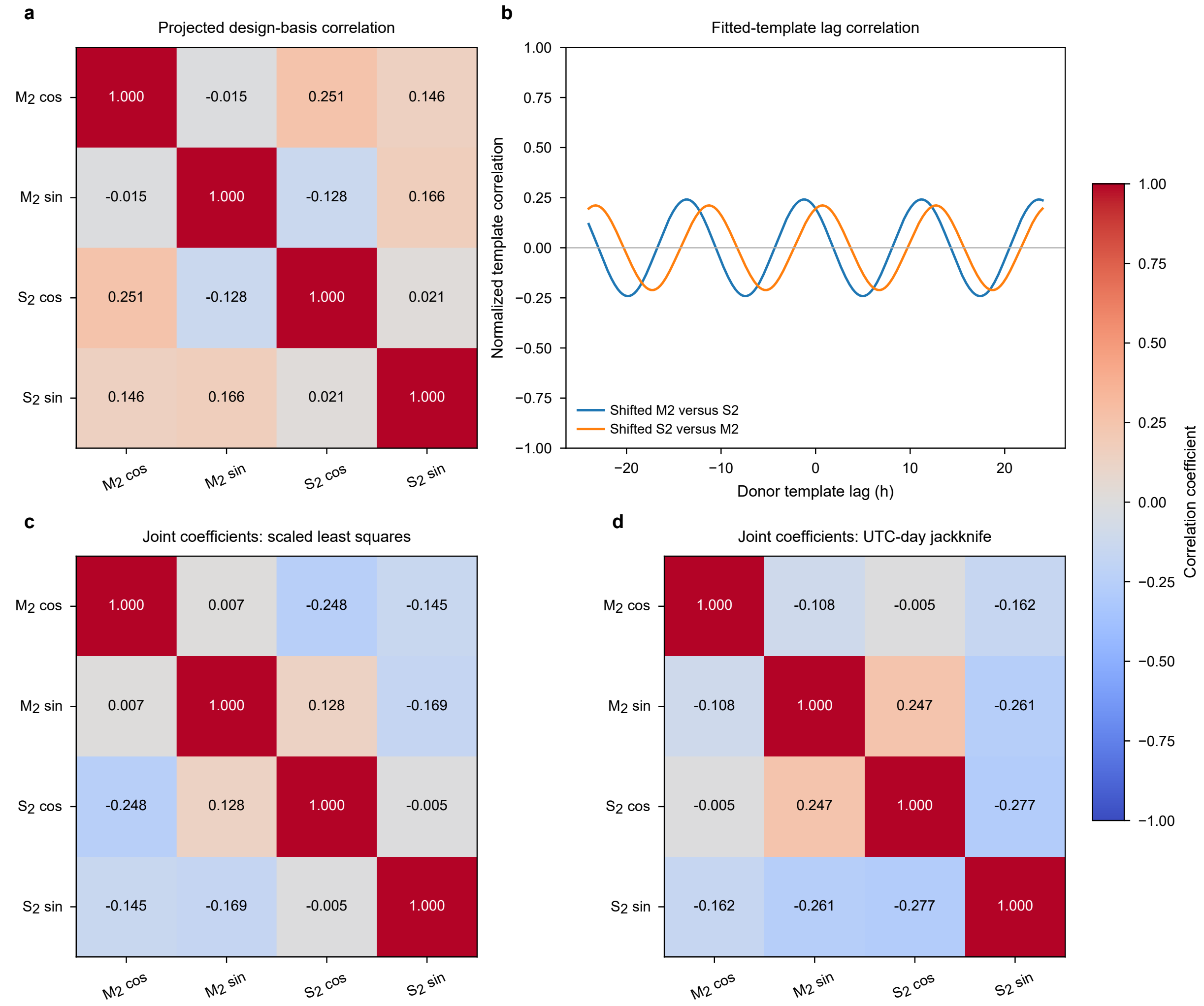
Extended Data Fig. 2 | Tidal records across campaigns and the April campaign. a, Three-hour averages of the filtered frequency ratio after subtraction of fitted segment offsets (blue), with the tidal model (red). Measurement-time axes compress gaps longer than 6 h, marked by vertical lines; gaps longer than 5 days are labelled. The vertical range is $\pm 15 \times 10^{-18}$; triangles mark values outside this range. **b,** Frequency-ratio observations after the 1,200-s triangular filter and removal of fitted segment offsets, their 3-h averages, and predictions with and without ocean loading. The horizontal axis retains actual calendar dates and gaps (local time, UTC+8). **c,** Sinusoidal power before and after subtracting the model, expressed as amplitude squared divided by two on a linear vertical axis, in units of 10^{-36}. The dashed curve is the 99% level at each frequency; the dotted line is the 99% level for the maximum over 15–50 μHz, both obtained from 1,000 resampled residual series. In the April record, K_1, P_1 and O_1 merge into a diurnal band and M_2, S_2 and N_2 into a semidiurnal band. Power below 8 μHz exceeds the displayed range.



Extended Data Fig. 3 | Amplitude spectra and sampling-response checks. All panels use the auxiliary reduction with the 3,299 recorded bin-centre times and their gaps, equal weights and 54 segment constants. Ordinates give equivalent-height peak amplitude in cm on linear scales, with 0–6 cm zooms below each panel. **a,** Observed amplitude spectrum and amplitude spectrum after subtraction of the fixed unit-amplitude physical prediction, as in Fig. 3c. **b,** Amplitude spectra before and after subtraction of the fitted M_2, S_2 or both waveforms, obtained from one simultaneous freely phased fit with N_2 and K_2 controlled. **c,** Local scans around M_2 conditioned on S_2, and around S_2 conditioned on M_2, with N_2 and K_2 controlled; grey is the baseline from b. **d,** M_2-only, S_2-only and combined fitted-waveform injections through the same sampling and projection, compared with the observed spectrum; injected amplitudes need not add. Dotted lines mark N_2, M_2, S_2 and K_2 in ascending frequency. Amplitudes at the controlled N_2 and K_2 frequencies are undefined, and responses close to them are weakly constrained.



Extended Data Fig. 4 | Amplitude and phase sensitivity to compensation. **a,c,** Subtracting a scaled M_2 waveform and re-estimating S_2; **b,d,** the reverse. **a,b,** Equivalent-height peak amplitude, converted as in Fig. 3c; **c,d,** phase relative to the common time origin. Solid curves use donor waveforms from the joint fit; dotted curves use single-frequency donor fits. All target-only fits retain segment offsets and free N_2/K_2 pairs. Dashed grey curves (donor pair also fitted) are invariant by construction. $\lambda = 1$ denotes subtraction of the estimated waveform. Amplitude axes are linear and share a scale across directions.



a

	M2 cos	M2 sin	S2 cos	S2 sin
M2 cos	1.000	-0.015	0.251	0.146
M2 sin	-0.015	1.000	-0.128	0.166
S2 cos	0.251	-0.128	1.000	0.021
S2 sin	0.146	0.166	0.021	1.000

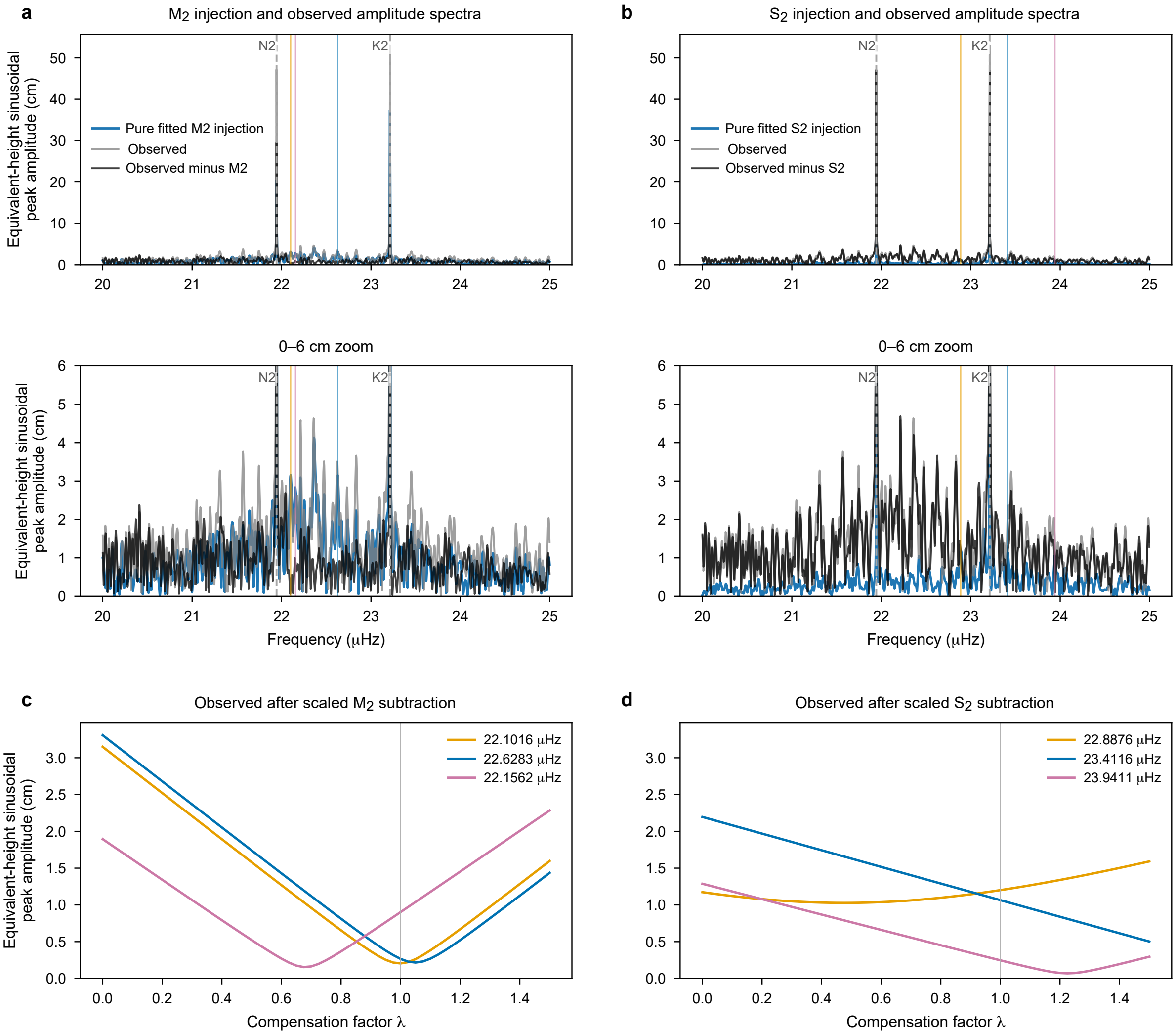
c

	M2 cos	M2 sin	S2 cos	S2 sin
M2 cos	1.000	0.007	-0.248	-0.145
M2 sin	0.007	1.000	0.128	-0.169
S2 cos	-0.248	0.128	1.000	-0.005
S2 sin	-0.145	-0.169	-0.005	1.000

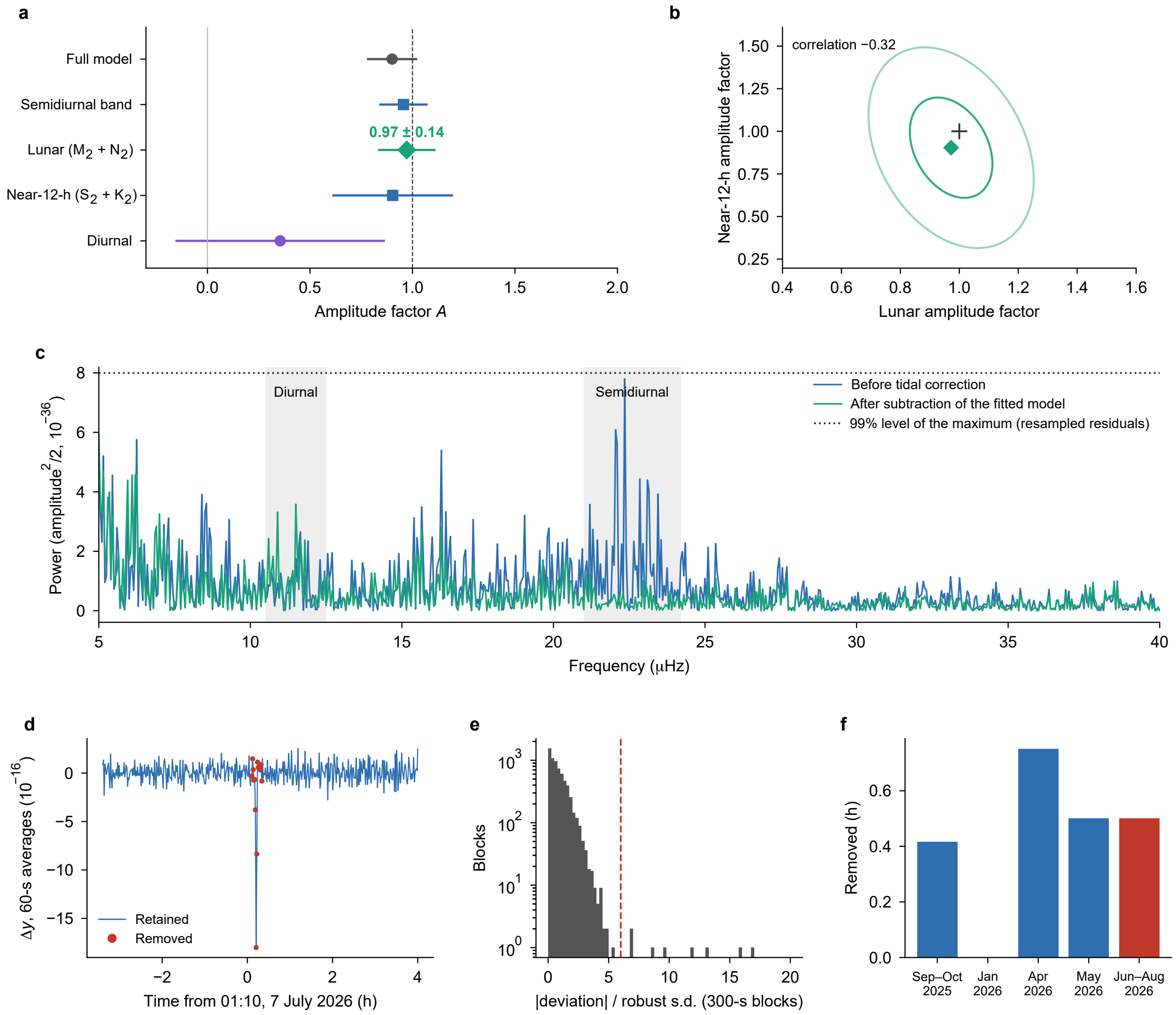
d

	M2 cos	M2 sin	S2 cos	S2 sin
M2 cos	1.000	-0.108	-0.005	-0.162
M2 sin	-0.108	1.000	0.247	-0.261
S2 cos	-0.005	0.247	1.000	-0.277
S2 sin	-0.162	-0.261	-0.277	1.000

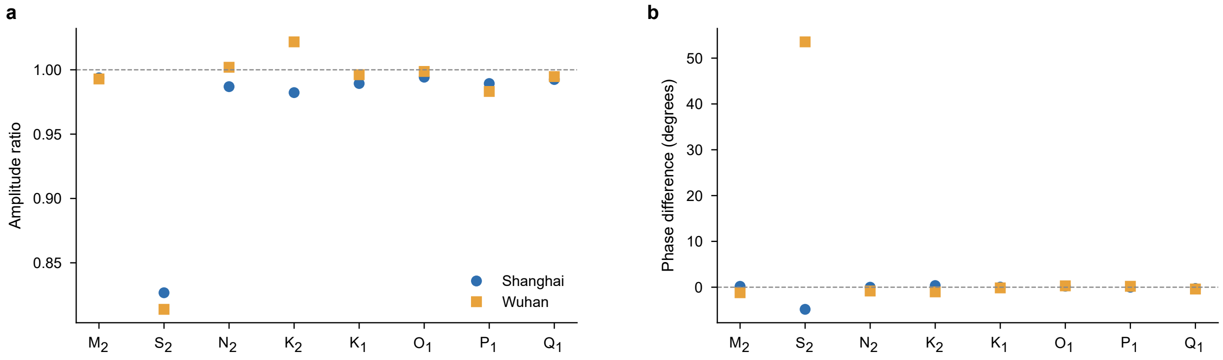
Extended Data Fig. 5 | Harmonic geometry and joint-coefficient correlations on the auxiliary 600-s record. a, Correlations of the M_2 and S_2 cosine and sine design columns after projecting out 54 segment constants and free N_2/K_2 pairs. b, Normalized lag correlations of the fitted templates, evaluated on the original timestamps without interpolating missing observations; these describe template similarity, not propagation delay. c,d, Correlations between the four jointly estimated coefficients (M_2 cosine, M_2 sine, S_2 cosine, S_2 sine) from the scaled least-squares covariance (c) and the UTC-day delete-one jackknife covariance (d). The three matrices share a −1 to +1 colour scale. The maximum canonical correlation of the projected M_2 and S_2 pairs is 0.290.



Extended Data Fig. 6 | Sampling aliases and amplitude response to compensation. a,b, Equivalent-height amplitude spectra of the observations, of the observations after subtraction of the jointly fitted donor waveform, and of that donor alone, on identical timestamps and nuisance projection, with 0–6 cm zooms below. Coloured vertical lines mark three off-centre candidate frequencies selected from the pure-injection response. **c,d,** Observed amplitude at those frequencies as the compensation factor varies. Amplitude axes are linear and share scales within rows. Amplitudes at the controlled N_2 and K_2 frequencies are undefined, and responses close to them are weakly constrained.



Extended Data Fig. 7 | Filtered-validation responses, spectra and sample selection. a, Amplitude factors for the full model and the semidiurnal band, each fitted with one amplitude, and for the lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal groups fitted jointly. All fits include segment offsets and omit empirical daily terms. Error bars are one standard deviation, calibrated by residual resampling for the full model, semidiurnal band and lunar group; for the near-12-h and diurnal groups, the larger of the scaled least-squares and day-by-day jackknife uncertainties is used. The dashed line denotes the prediction, $A = 1$. **b,** Covariance contours at radii one and two for the lunar and near-12-h amplitudes, using the least-squares correlation scaled to the quoted marginal uncertainties; the cross denotes the model prediction. **c,** Sinusoidal power of the full record after projection of segment offsets, before and after subtracting the fitted full model. Power is the squared sinusoidal amplitude divided by two, on a linear axis in units of 10^{-36}. The dotted line is the 99% quantile of the maximum power over 5–40 μHz from 200 resampled residual series. **d,** Sixty-second averages around the excursion on 7 July 2026; red points mark averages consisting mainly of samples excluded by the criterion in Methods. **e,** Distribution of deviations of 300-s means from a running median over ±30 min, divided by the robust standard deviation within each segment. The dashed line marks the threshold of six. The count axis is logarithmic. **f,** Observation time excluded in each campaign (summer in red).



Extended Data Fig. 8 | Radial-displacement validation of the loading convolution. a,b, Amplitude ratios (**a**) and phase differences (**b**) between radial displacements calculated with the loading Green's function and the FES2022 load-tide grid in the centre-of-mass frame, at Shanghai and Wuhan, for eight constituents. The S_2 difference is discussed in Supplementary Methods SM1.

a | Main and filtered validation analyses.

	Filtered validation (primary)	Main analysis (auxiliary)
Data	1-s samples retained after the filtered-validation selection and transient screening	3,299 non-overlapping 600-s means after segment-median counter-jump rejection and per-segment wall-clock binning (Methods), without the additional running-median transient screening; 545.9 h
Filtering	1,200-s triangular window at 600-s steps, applied to data and model	None
Tidal template	IERS 2010 solid tide with constituent-dependent Love numbers; ocean-loading potential by Green's-function convolution of FES2022 (SM1)	IERS radial displacement converted with h_2 and k_2; effective-potential loading convolution with 18 constituents (Methods)
Weights	Equal	Observing-period noise for the full-template and three-group fits; equal for spectral scans, joint sinusoid fits and the solid-Earth/loading fit
Uncertainties	Residual resampling for the full template, semidiurnal band and lunar group; larger of scaled least squares and day-by-day jackknife otherwise	Larger of scaled least squares and UTC-day jackknife
Used in	Fig. 1b,d, Figs. 2 and 4, Extended Data Figs. 1, 2 and 7, Extended Data Tables 1a,b and 2a,b	Fig. 3, Table 1, Extended Data Figs. 3–6, Extended Data Table 2c,d

b | Injection tests of the primary estimator.

Disturbance	A_{true}	Bias	s.d.	Median error	1σ coverage	95% binomial interval	1.96σ coverage	$A > 3\sigma$
None	0	−0.026	0.134	0.135	0.70	0.645–0.751	0.94	0.003
None	1	−0.003	0.135	0.137	0.71	0.659–0.764	0.94	1.000
24 h, air-temperature phase	0	−0.013	0.141	0.136	0.61	0.549–0.662	0.93	0.000
24 h, air-temperature phase	1	−0.007	0.145	0.138	0.64	0.579–0.691	0.94	1.000
Common 24 h + 12 h	0	−0.013	0.148	0.140	0.63	0.573–0.685	0.90	0.000
Common 24 h + 12 h	1	−0.024	0.148	0.139	0.60	0.546–0.659	0.92	1.000
Summer 24 h + 12 h	0	−0.021	0.138	0.140	0.67	0.610–0.720	0.95	0.000
Summer 24 h + 12 h	1	−0.016	0.150	0.137	0.59	0.529–0.643	0.92	1.000

c | Noise-model simulations of the alternative processing.

Noise model	Bias (T)	Coverage (T), %	Bias (L)	Coverage (L), %
White	−0.000	96.7	−0.001	98.3
White + OU, 1 h	−0.007	94.5	−0.002	96.9
White + OU, 12 h	−0.002	96.1	+0.031	98.8
White + OU, 3 d	+0.003	95.9	+0.004	99.8
White + period constants	+0.001	96.0	+0.004	98.3
Student-t_7 MA(1)	+0.002	98.3	+0.004	98.2
Student-t_5 MA(1)	+0.001	98.9	+0.003	98.3
Student-t_7 MA(1) + OU	+0.001	97.2	+0.005	98.5

Extended Data Table 1 | Analysis definitions and uncertainty checks.

a, Main and filtered validation analyses. Both reductions use the same 54 observing segments, each with its own offset. They differ in selection, filtering, tidal-template realization and weighting, so their estimates are related but are not independent measurements.

b, Injection tests of the primary estimator. Lunar amplitudes fitted with segment offsets and no empirical daily terms to resampled residuals plus either the full tidal model ($A_{\text{true}} = 1$) or the model without the lunar group ($A_{\text{true}} = 0$). The four daily-disturbance scenarios are defined in Methods, with 300 realizations per row. Bias and standard deviation describe the fitted amplitude. The median standard error is that computed within each realization, before the final residual-based calibration. Coverage is the fraction of intervals, formed with these uncalibrated standard errors, containing the injected amplitude; with the calibrated uncertainty of 0.14 the one-standard-error coverage is 0.67–0.68 (Supplementary Methods SM3); the accompanying 95% binomial interval refers to one-standard-error coverage. The last column gives the fraction with $A > 3\sigma$.

c, Noise-model simulations of the alternative processing. Bias and coverage of nominal 95% intervals based on the largest of the least-squares, delete-day and delete-period standard errors. T denotes the full template without daily terms, and L the lunar group with daily terms, using the ten-day calendar-baseline design in Extended Data Table 2b and the actual observation times. Gaussian scenarios contain 1,024 simulations; Student-t and moving-average scenarios contain 4,096. OU denotes Ornstein–Uhlenbeck noise and MA(1) a first-order moving-average process.

a | Daily terms and campaign-specific tidal amplitudes.

Campaign	Days	24-h amplitude	24-h maximum, h	12-h amplitude	A	A_{lunar}
Sep–Oct 2025	4	5.1 ± 2.7	20.7 ± 5.6	5.0 ± 9.7	1.28 ± 0.31	0.04 ± 0.74
Jan 2026	4	2.1 ± 3.0	17.2 ± 3.9	4.1 ± 1.8	1.27 ± 0.30	1.36 ± 0.37
Apr 2026	5	2.9 ± 2.2	1.9 ± 1.4	1.0 ± 3.9	0.91 ± 0.13	0.66 ± 0.42
May 2026	4	5.2 ± 3.3	3.8 ± 2.4	3.1 ± 2.6	0.97 ± 0.18	0.28 ± 0.68
Jun–Aug 2026	22	2.8 ± 1.3	15.8 ± 1.5	0.6 ± 0.7	0.67 ± 0.18	0.87 ± 0.18

b | Related tidal estimates.

Analysis	Data and model	Amplitude factor A	Uncertainty
Hou et al.[23]	Summer; 16 segments; their tidal model and segment-wise correlation	0.87(15)	Reported weighted statistical error
Primary reduction, lunar group	54 segments; lunar group; segment offsets; astronomical ephemerides	0.97 ± 0.14	Calibrated by residual resampling
Primary reduction, full template	Same observations; full template; segment offsets	0.90 ± 0.12	Calibrated by residual resampling
Alternative processing	2026 observations; 600-s means; ten-day calendar baselines; full template	0.88 ± 0.10	Largest of least-squares, delete-day and delete-period standard errors
Alternative processing	Same observations and baselines; lunar group; daily terms in each ten-day block	0.89 ± 0.28	Largest of least-squares, delete-day and delete-period standard errors

c | Free sinusoidal coefficients in the auxiliary reduction.

Coefficient	Estimate	Scaled least-squares standard error	UTC-day jackknife standard error
M_2 cosine	-1.667	0.688	0.880
M_2 sine	-4.185	0.679	0.635
S_2 cosine	-0.460	0.738	0.701
S_2 sine	-1.866	0.750	1.043

d | Template-response signal-to-noise ratios.

Fit and coefficient	$\hat{A}_i \pm u_i$	ρ_i	$\rho_{i,\mathrm{white}}$	η_i
Full tide, one template	0.955 ± 0.108	8.83	9.79	0.865
Lunar group, three-group fit	1.000 ± 0.132	7.58	7.50	0.840
Near-12-h group (S_2, K_2)	0.925 ± 0.365	2.54	3.14	0.856
Diurnal group, three-group fit	0.579 ± 0.538	1.08	1.55	0.740
Solid Earth, two-template fit	1.082 ± 0.322	3.36	5.81	0.611
Ocean loading, two-template fit	0.486 ± 0.544	0.89	1.17	0.612

Extended Data Table 2 | Response estimates and related analyses.

a, Daily terms and campaign-specific tidal amplitudes. Amplitude and local time of maximum of the 24-h term, and amplitude of the 12-h term, fitted jointly with the full tidal model within each campaign. Days are local calendar days with data. Daily amplitudes are expressed in 10^{-18}. Their uncertainties are the larger of the scaled least-squares and day-by-day jackknife values; uncertainties in the time of maximum are jackknife estimates. Weak sinusoidal amplitudes are biased positive. The full-template amplitude A, fitted without daily terms, and its residual-calibrated uncertainty are those in Extended Data Fig. 1a. The last column gives the lunar amplitude fitted independently to each campaign with the primary estimator.

b, Related tidal estimates. These estimates share observations and are not combined. The last column specifies each uncertainty definition. Applying the segment-wise estimator of Hou et al.[23] to the summer record with the tidal model used here gives 0.78 ± 0.19.

c, Free sinusoidal coefficients in the auxiliary reduction. Joint M_2 and S_2 cosine–sine coefficients, after controlling 54 segment constants and freely phased N_2 and K_2 terms. Coefficients and standard errors are in units of 10^{-18}. Phases refer to 6 September 2025, 16:05 UTC. Scaled least-squares errors use the residual variance; UTC-day jackknife errors delete each of the 43 observed dates and refit all coefficients.

d, Template-response signal-to-noise ratios. The first four rows use the non-overlapping full-template or three-group fits; their empirical errors are the larger of scaled-least-squares and UTC-day-jackknife errors. The final two rows use the equal-weight complete solid-Earth/loading fit with UTC-day-jackknife errors. The common-white-noise calculation retains each estimator and uses $\alpha = 5.4 \times 10^{-16}\sqrt{\mathrm{s}}$. The dimensionless factor η_i is defined in Methods.

Supplementary Information

Tidal potential measured with distant optical clocks

Lei Hou et al.

Reference numbers follow the main manuscript.

Supplementary Methods

SM1–SM5 describe the filtered validation analysis, also labelled the primary reduction. SM6 and SM7 describe the main analysis, also labelled the auxiliary reduction, unless a filtered-data comparison is explicitly identified. Extended Data Table 1a summarizes their methodological differences.

SM1. Physical model

We use $k_{20} = 0.29525$, $k_{22} = 0.29801$ and $k_3 = 0.093$. The IERS 2010 resonance formula, applied constituent by constituent for k_{21}, gives 0.2588 for K_1, 0.2873 for P_1 and 0.2975 for O_1. The radial displacement is computed with the IERS algorithm, including the frequency and latitude dependence of the Love number h [1].

PREM load Love numbers are used through degree $n = 696$ [32], from the gravity-toolkit distribution [33]. Above this degree, h'_n is joined smoothly to its Boussinesq limit of −6.22, and the series is summed with a Kummer transformation through $n = 40,000$. The convolution uses full grid resolution within 3° of each station and sums blocks of 8×8 cells farther away, which changes the differential M_2 and K_1 loading by at most 0.05%.

The full loading potential exceeds the vertical-displacement contribution $-g\delta h$ by factors of approximately 1.4 for the semidiurnal and 1.8 for the diurnal constituents. The radial displacement from the same convolution agrees with the FES2022 load-tide grid to within 2.2% in amplitude and 1.2° in phase for seven of the eight constituents (Extended Data Fig. 8); the attraction and induced-potential terms use the same PREM load Love numbers. For S_2, whose load-tide grid derives from a different ocean-model release, the amplitude differs by about 18% and the phase at Wuhan by 54°. The S_2 loading is 0.62×10^{-18} in Shanghai and 0.18×10^{-18} in Wuhan, a differential amplitude of 0.63×10^{-18} compared with 1.42×10^{-18} for M_2. Substituting the S_2 loading inferred from the load-tide grid changes the full-template amplitude by 0.001 and the semidiurnal amplitude by 0.000, and leaves the lunar amplitude unchanged.

The Shanghai clock is 22 km from the nearest ocean cell. Filling land cells within 0.3° or 0.6° of the coastline in a $4^\circ \times 5^\circ$ box centred on Shanghai with the nearest ocean values, strips approximately

33–66 km wide, increases the Shanghai M_2 loading potential expressed as $\delta W/g$ from 15.3 mm to 17.0 or 18.5 mm and decreases the lunar response by 0.05 or 0.08, respectively.

Setting the remainder of the eight-constituent model to zero or doubling it gives semidiurnal amplitudes of 0.95–0.96 and lunar amplitudes of 0.95–0.99.

SM2. Selection and timing

Adding the tidal model with factors of −1, 1 or 3 leaves the removed samples unchanged; six robust standard deviations is the lowest tested threshold with this invariance up to three times the model. Block lengths of 60 or 300 s and thresholds from four to eight standard deviations give lunar amplitudes from 0.93 to 1.02. Retaining the excursions gives 0.96 ± 0.14, and the full-template and semidiurnal amplitudes change by less than 0.04. Before transient removal, the procedure reproduces the summer sample set of ref. 23; after removal, the campaign diurnal amplitudes are mutually consistent ($p = 0.21$). Three of the nine identified excursions coincide with approximately equal and opposite link-monitor excursions.

Omitting endpoint trimming changes the lunar amplitude by −0.009 and the full-template amplitude by 0.008. Shifting the model by ±10 min changes A by less than 0.02. Long-period tides are not fitted: the similar site latitudes make the predicted differential semiannual tide only about 2×10^{-20}. Durations refer to retained 1-s samples; the exposure of the filtered data at 600-s steps is slightly shorter.

SM3. Uncertainty calibration and campaign consistency

For the lunar group, two independent sets of 200 realizations give standard deviations of 0.13 and 0.14, compared with 0.13 for the larger of the scaled least-squares and jackknife errors; we quote 0.14, with a resampling bias of +0.01. With the daily disturbances of Methods, which have r.m.s. values up to 1.9×10^{-18}, 5,000 realizations per scenario give standard deviations rounding to 0.14 (0.13–0.15 for the 300 realizations of Extended Data Table 1b) and coverage of 0.67–0.68 for ±0.14 and 0.945–0.952 for ±1.96 × 0.14.

For the semidiurnal response, the group-bootstrap standard deviation of 0.08 is smaller than the calibrated value of 0.12, which is quoted. The segment bootstrap gives a lunar amplitude of $0.98 \pm$

0.11 and phase of $-6^\circ \pm 8^\circ$. The correlation between the lunar and neighbouring semidiurnal coefficients is −0.32.

Summer supplies 61% of the conditional information on the lunar response and gives 0.87 ± 0.18, compared with 1.05 ± 0.31 for the earlier campaigns together. The five lunar estimates are consistent ($\chi^2 = 4.1$, four degrees of freedom, $p = 0.40$; Extended Data Table 2a), as are the semidiurnal estimates ($\chi^2 = 3.2$, $p = 0.52$; Extended Data Fig. 1a,b). Resampling the operating groups gives a 95% lunar interval of [0.76, 1.18].

SM4. Prediction, environmental and daily-term checks

The whole-template prediction reduces residual sums of squares in seven of eight held-out periods, by 4.9% when pooled; the lunar prediction improves six of eight periods, by 1.7% when pooled.

Over the 98-h overlap, the blackbody shift calculated from chamber temperature and filtered like the ratio has an r.m.s. of 2.8×10^{-18}. As a free regressor it gives 0.92 ± 0.74, or 0.72 ± 0.74 after excluding the chamber-temperature transient on 30 June (day-by-day jackknife errors), consistent with zero. Without real-time correction, the shift would contain a daily component of 2.0×10^{-18} peaking at 12:13 local time, with a peak-to-peak daily range of 4×10^{-18}; the fitted 24-h ratio term in the same overlap is $(0.9 \pm 1.1) \times 10^{-18}$ with maximum at 10:54 ± 5.0 h.

On the primary sampling after segment-mean removal, the predicted solid-tide and loading r.m.s. fractional frequencies are 3.02×10^{-18} and 1.35×10^{-18}, and that of their sum is 4.09×10^{-18}. The response fitted to the solid-Earth tide alone is 1.19 ± 0.20, compared with 1.32 expected on this sampling for the combined model, and the separately fitted loading coefficient is 0.89 ± 0.55. This coefficient comes from a fit of the solid-Earth and loading templates with two free coefficients and segment offsets to the filtered primary data with equal weights; its uncertainty is the larger of the scaled least-squares and day-by-day jackknife errors.

The summer 24-h term has amplitude $(2.8 \pm 1.3) \times 10^{-18}$ and peaks at 15:48 local time with a 1.5-h uncertainty; the 12-h term is $(0.6 \pm 0.7) \times 10^{-18}$. With summer projection coefficients of 0.016 and 0.043 per 10^{-18}, they shift the lunar response by at most 0.045 and 0.024 (Extended Data Table 2a). For the full record, a common-phase 12-h perturbation of 10^{-18} changes the whole-template response by at most 0.087.

The lunar responses for alternative daily models (Fig. 4a) are: campaign-specific 24-h terms with air-temperature phases, 0.99 ± 0.14; free-phase campaign-specific 24-h terms, 0.97 ± 0.15; air-temperature anomalies at both sites, 0.98 ± 0.13; common 24-h and 12-h terms, 1.00 ± 0.13; campaign-specific free-phase 12-h terms, 0.82 ± 0.17, or 0.86 ± 0.18 with 24-h terms also included. Operating-group or period offsets in place of segment offsets give 0.97 ± 0.11 and 0.99 ± 0.11, and omitting each earlier campaign in turn gives 0.89–0.99.

SM5. Alternative processing and spectra

The segment-wise estimator of ref. 23 demeans filtered data and model within each segment, fits the amplitude through the origin using non-overlapping 1,200-s windows, and combines segments with inverse-variance weights. For the summer record it gives $A = 0.78 \pm 0.19$ with the present physical model and 0.87(15) with the model of ref. 23; the joint lunar fit gives 0.87 ± 0.18.

Independent processing of the 2026 records uses non-overlapping 600-s means, corrected clock systematic shifts and fixed ten-day calendar baselines. It gives a whole-template response of 0.88 ± 0.10 without empirical daily terms and a lunar response of 0.89 ± 0.28 with daily terms in each ten-day block, and its whole-template prediction improves all six held-out 2026 periods. Simulations with white, Ornstein–Uhlenbeck (correlation times of 1 h to 3 d), Student-t and moving-average noise give biases below 0.04 and 95%-interval coverage of 94.5–99.8% (Extended Data Table 1c). Simpler solid-tide approximations change its whole-template scale by less than 0.01.

At M_2, the harmonic power of the full record after projection of segment offsets decreases by 99% after subtraction of the fitted whole template (Extended Data Fig. 7c). In the three-group fit the diurnal coefficient is 0.35 ± 0.51; the two-band estimate is 0.36 ± 0.50 (Extended Data Fig. 1b).

SM6. Auxiliary response and uncertainty details

Extended Data Table 1a compares the main and filtered validation analyses; Extended Data Table 2d gives the main response estimates and their uncertainties. In the phase-relaxed fit, the lunar in-phase and quadrature estimates are 1.0007 ± 0.1320 and -0.0969 ± 0.1989; the scaled least-squares quadrature error is 0.1273.

The projected solid-Earth/loading template correlation is 0.725 and the coefficient correlation under day deletion is −0.888. For M_2 and S_2, at the measured amplitudes 4.5047×10^{-18} and 1.9214×10^{-18}, observing-period noise propagation gives standard errors of 0.6153×10^{-18} and 0.6788×10^{-18}. For the diurnal-group and loading coefficients, the period-noise standard errors are 0.3631 and 0.4337 and the UTC-day-jackknife errors 0.5381 and 0.5438.

Propagating the observing-period noise coefficients, $\Sigma_{kk} = \alpha^2_{p(k)}/\Delta t_k$, through the unchanged estimators gives S/R values of 7.32, 2.83, 1.59 and 1.12 for M_2, S_2, the diurnal group and ocean loading. In the triangular-filter reduction, the corresponding ratios are 6.93 for the lunar group, 3.10 for the neighbouring semidiurnal group and 0.72 for the two-band diurnal estimate.

SM7. Compensation, alias and correlation diagnostics

For donor D and target T, the joint fit supplies the fixed donor coefficient vector $\widehat{\beta}_D$. For 61 values of λ from 0 to 1.5, we form $y_\lambda = y - \lambda X_D \widehat{\beta}_D$ and fit only the target pair together with N. With $P_T = R_N X_T$ and $P_D = R_N X_D$,

$$\widehat{\beta}_T(\lambda) = \widehat{\beta}_{T,\text{single}} - \lambda P_T^+ P_D \widehat{\beta}_D,$$

where P_T^+ is the Moore–Penrose pseudoinverse. We repeat the sweep with the donor's single-frequency estimate. A fit retaining both pairs after subtraction leaves the target unchanged by construction and is an algebraic control. Amplitude is the norm of the two coefficients and phase is atan2$(\beta_{\text{sin}}, \beta_{\text{cos}})$, corresponding to $A\cos(2\pi f(t-t_0)-\phi)$. Amplitudes are converted to equivalent height as in Methods.

Template lag correlations evaluate the fitted donor waveform at $t-\ell$ and the target at t, residualize both on the observation support and normalize their inner product, for lags within ±24 h in 5-min steps. The zero-lag correlation is 0.1955, and the maximum absolute lag correlations are 0.2413 for shifted M_2 against S_2 and 0.2113 in the reverse direction. The largest cross-component coefficient correlations between the joint M_2 and S_2 estimates, $R_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$, are 0.248 under scaled least squares and 0.261 under UTC-day deletion (Extended Data Fig. 5).

For alias diagnostics, we scan 20–25 μHz with spacing $1/(12T_{\text{span}})$ and, for each donor, select the three strongest local maxima of its pure-injection least-squares power at least $2/T_{\text{span}}$ from its own frequency. At each selected frequency, the fitted coefficient vector follows $\hat{b}_f(\lambda) = \hat{b}_f(0) - \lambda\hat{b}_{f,D}$, and the equivalent-height amplitude is $K\|\hat{b}_f(0) - \lambda\hat{b}_{f,D}\|$ with $K = (c^2/g_0)\,10^{-16}$ cm. At 22.1016 and 22.6283 μHz, subtraction of the jointly fitted lunar waveform reduces the amplitudes from 3.149 to 0.204 cm and from 3.307 to 0.266 cm, reductions of 93.51% and 91.97%, with minima at compensation factors 0.9974 and 1.0490 (Extended Data Fig. 6).